

LESSON 2-9 • SESSION 2 • PRACTICE • FORM B

Comparing Data Sets Using MAD

Learning target – I can compare two data sets that share a mean by comparing their mean absolute deviations. · 6.SP.B.5c

REMEMBER

- Two sets can share a mean and still have different MADs. The **mean** is the center; the **MAD** is the spread.
- The set with the **smaller MAD** is steadier (more consistent). If the MADs are given, just compare them.

WORKED EXAMPLE

Baker A made 8, 9, 11, 12 cookies per tray. Baker B made 4, 8, 12, 16. Both means are 10. Who is steadier?

Step 1 A's distances: 2, 1, 1, 2. $MAD = 6 \div 4 = 1.5$.

Step 2 B's distances: 6, 2, 2, 6. $MAD = 16 \div 4 = 4$.

Step 3 $1.5 < 4$, so Baker A is steadier.

Answer Baker A

WORDS **variation** (*variación*) · **consistent** (*consistente*) · **mean absolute deviation** (*desviación media absoluta*)

PRACTICE

show your work · label units

1. **Snack Sales.** Both stands have a mean of 64. **Stand 1:** 58, 62, 66, 70. **Stand 2:** 46, 60, 68, 82.

a. Stand 1 MAD: _____ b. Stand 2 MAD: _____ c. Steadier: _____

2. **Two Trails.** Both means are 30 min. **Trail A:** 22, 27, 33, 38. **Trail B:** 25, 28, 32, 35.

Hint Work one trail at a time: distances first, then average.

- a. Trail A MAD: _____
 b. Trail B MAD: _____
 c. More predictable: _____

Show your work.

3. **Same MAD.** Set G has mean 80 and MAD 4. Set H has mean 40 and MAD 4. Priya says the sets are identical. Is she right?

Hint The mean is the center; the MAD is the spread.

Show your work.

Answer _____

4. **Picking a Team.** Team P averages 75 points with MAD 12. Team Q averages 75 points with MAD 2. Which team is more reliable?

Show your work.

Team _____

5. **Garden Beds.** Both means are 49 cm. **Bed 1:** 42, 46, 52, 56. **Bed 2:** 33, 45, 53, 65.

- a. Bed 1 MAD: _____
 b. Bed 2 MAD: _____
 c. More even: _____

Show your work.

6. **Same Mean, Same MAD.** Set J is 32, 36, 44, 48. Set K is 34, 34, 46, 46. Both means are 40.

Hint Find each MAD on its own, then look at the values.

- a. MAD of J: _____
 b. MAD of K: _____
 c. Are the sets the same? Explain. _____

Show your work.

7. **Choose a Supplier.** Supplier 1: mean 6 days, MAD 0.5 day. Supplier 2: mean 5 days, MAD 2.5 days. Which would you pick to meet a deadline? Why?

Hint Weigh the faster average against how reliable each is.

Show your work.

Supplier _____

8. **Two Archers.** Both archers have a mean score of 7. Archer 1: 6, 8, 8, 6. Archer 2: 3, 10, 4, 11.

- a. Archer 1 MAD: _____
 b. Archer 2 MAD: _____
 c. Pick an archer for a tie-breaking shot. Explain. _____

Show your work.

FIND THE ERROR

explain the mistake, then fix it

9. **Mateo's Work.** Mateo wrote: "Set R has the larger mean, so Set R must have the larger MAD." Explain his mistake.

TEST PRACTICE

choose the best answer

10. Equal means and unequal MADs mean the sets differ in –
A center **B** spread **C** size **D** total
11. For data that is roughly symmetric, the best pair of measures is –
A median and IQR **B** range and mode **C** mean and MAD **D** median and range

SELF-CHECK CODE

find your answer · copy its letter into that problem's box · spell a math word

14 Z	1 E	5.5 S	10 U	44 Q	11 B	5 L	3.5 O	6 T	4 A
#1a	#1b	#2a	#2b	#5a	#5b	#6a	#8a		

CHECK YOURSELF

☐ I can do this on my own ☐ I need a little help ☐ I have a question about #____